Dongwook **Kwon**

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Experience

Suresoft Technologies — *Seongnam, Gyeonggi, Korea*
AI Engineer, Intelligent Agent Technology Team — *Dec. 2025 – Present*
- Developing an AI validation system utilizing AI agent–based architectures.
- Designing and implementing systems to verify and evaluate AI models through autonomous and collaborative AI agents.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University — *Seoul, Korea*
Research Assistant — *Aug. 2021 – Dec. 2025*
- Conducted research on deep learning algorithms, focusing on anomaly detection in time-series data, including Electrocardiogram (ECG) and Cyber-Physical Systems (CPS).
- Participated in the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada — *Toronto, ON, Canada*
Research Project — *Jan. 2025 – Jul. 2025*
- Participating in the development of an assessment framework for agentic AI systems utilizing LLMs to evaluate performance.

Qualcomm Institute, University of California San Diego — *San Diego, CA, USA*
AI Development Project, Summer Program — *Jul. 2022 – Aug. 2022*
- Participated in the Personality-Based Drug Addiction Prediction System project.
- Attended seminars on deep learning and machine learning algorithms.

Education

Kwangwoon University — *Seoul, Korea*
Master of Science in Computer Engineering — *Mar. 2024 – Feb. 2026*
- GPA: 4.33/4.5 (Percentage: 98.1%)

University of Toronto — *Toronto, ON, Canada*
Graduate Research Program in Mechanical and Industrial Engineering — *Jan. 2025 – Jul. 2025*
- GPA: 3.68/4.0

Kwangwoon University — *Seoul, Korea*
Bachelor of Science in Electronics and Communications Engineering — *Mar. 2019 – Feb. 2024*
- GPA: 3.41/4.5 (Percentage: 88.1%)

Honors & Awards

Major Awards

- *Aug. 2022* **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego — *United States*
- *Dec. 2021* **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project – Minister of Oceans and Fisheries — *South Korea*

Academic Recognition

- *Nov. 2025* **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers — *South Korea*
- *Nov. 2024* **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) — *South Korea*
- *Nov. 2023* **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering — *South Korea*

Competitions

- *Dec. 2025* **Silver Award**, 23rd Embedded Software Competition (KIISE) — *South Korea*
- *Dec. 2022* **Bronze Award**, Hanium ICT Mentoring Competition – FKII — *South Korea*
- *Sep. 2022* **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) — Kwangwoon University — *South Korea*
- *Oct. 2021* **Grand Prize**, MY (Multi-Y) Capstone Design Competition — Kwangwoon University — *South Korea*

Projects

Agentic AI Evaluation Framework — *Toronto, ON, Canada*
LG Electronics Canada — *Jan. 2025 – Present*
- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

AI-based Intrusion Detection and Attack Packet Crafting Technologies for APR1400 — *Seoul, Korea*
Sejong University — *Jul. 2022 – Present*
- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Quadruped Robot Using Vision SLAM with a Depth Camera — *Seoul, Korea*
Ministry of Science and ICT — *Apr. 2022 – Nov. 2022*
- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Active Kill-Switch and Biomarker-Based Defense for Life-Threatening IoT Medical Devices — *Seoul, Korea*
Kookmin University — *Sep. 2021 – Dec. 2022*
- Developed a security system for heart implant devices using an anomaly detection algorithm.

Development of a Smart Logistics System Using Computer Vision — *Seoul, Korea*
Ministry of Oceans and Fisheries — *Apr. 2021 – Nov. 2021*
- Developed a logistics detection system incorporating location tracking and stack layer identification for containers.

Publications

Journal Articles
- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, “Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise,” *Electronics*, vol. 14, no. 5, p. 890, 2025.
- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments,” under review at *Data Mining and Knowledge Discovery* (Springer Nature), 2025.

Conference Papers
- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades,” in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) – Industry Track (Poster)*, 2026. [†]Equal contribution
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, “GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems,” in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. [†]Equal contribution
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, “Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers,” in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. — **Best Poster Award – Silver Winner**
- **D. Kwon**, Y. Kang, and C. Park, “Enhancing medical device security with GNN-GRU anomaly detection model,” in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. — **Outstanding Paper Award**

Patents

- **Issued Patent**, “Method for Implantable Medical Device to Detect Anomaly in Real Time,” Patent No. KR 10-2961984, Issued on April 30, 2026.
- **Issued Patent**, “GPT API-Based Network Anomaly Detection,” Patent No. KR 10-2848814, Issued on August 18, 2025.

Leadership Experience

- **President**, Kwangwoon International Student Association — Lead intercultural programs and student engagement initiatives.
- **Director of Filming and Editing**, Kwangwoon Broadcasting Center (KWBC) — Oversaw media production and content delivery.
- **Class Representative**, Dept. of Electronics and Communications Engineering — Represented student interests in academic affairs.
- **Executive Member**, Amateur Radio Club — Organized radio workshops and technical exhibitions.